SYSTEM INDEX

A1 | Bare PCB + hand soldering | Existing assembled modules remain external

PROJECT STRUCTURE

02 01 Power distribution and ADC

文件: SmartBin_A_01_01.kicad_sch

03 02 MCU, MCP23017 and emergency stop

文件: SmartBin_A_02_01.kicad_sch

04 03 Serial module interfaces

文件: SmartBin_A_03_01.kicad_sch

05 04 Buffered outputs and actuators

文件: SmartBin_A_04_01.kicad_sch

06 05 Sensor and user inputs

文件: SmartBin_A_05_01.kicad_sch

Assembly guidance and external module connections are specified in the accompanying manual.

Do not connect the core 3.3 V rail in parallel with the external 3.3 V regulator.

Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

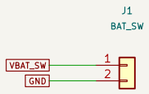
Sheet: /
File: SmartBin_A.kicad_sch

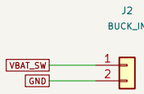
Title: SmartBin A / System index

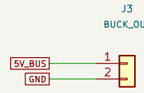
Size: A3	Date: 2026-10-12	**Rev: A1**
KiCad E.D.A. 10.0.6		Id: 1/6

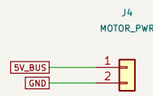
01 POWER DISTRIBUTION AND ADC

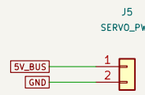
A1 | Bare PCB + hand soldering | Existing assembled modules remain external

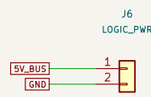




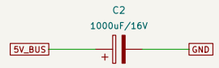


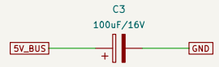


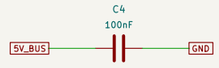


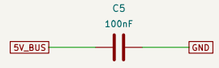


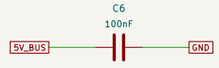








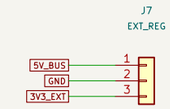


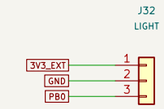


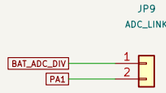












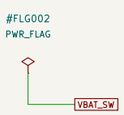


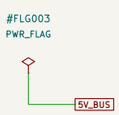


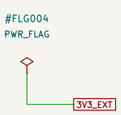


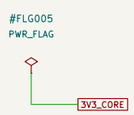












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.

STM32 + RFID / DIY through-hole carrier

Sheet: /02 01 Power distribution and ADC/
File: SmartBin_A_01_01.kicad_sch

Title: SmartBin A / 01 Power distribution and ADC

Size: A3	Date: 2026-10-12	**Rev: A1**
KiCad E.D.A. 10.0.6		Id: 2/6

02 MCU, MCP23017 AND EMERGENCY STOP

A1 | Bare PCB + hand soldering | Existing assembled modules remain external









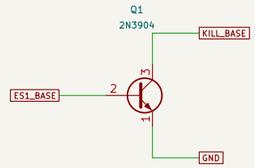


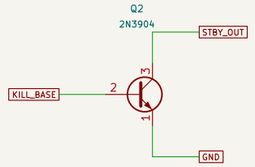


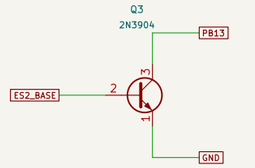


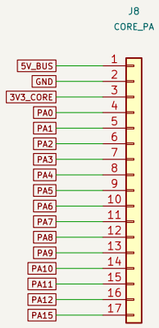


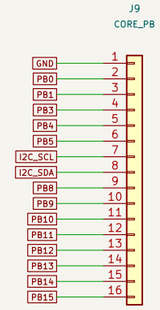


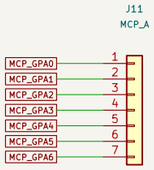


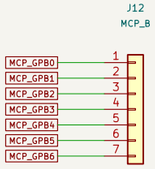


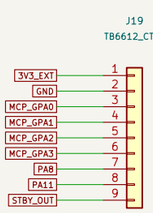


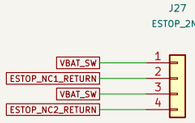












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

Sheet: /03 02 MCU, MCP23017 and emergency stop/
File: SmartBin_A_02_01.kicad_sch

Title: SmartBin A / 02 MCU, MCP23017 and emergency stop

Size: A3	Date: 2026-10-12	**Rev: A1**
KiCad E.D.A. 10.0.6		Id: 3/6

03 SERIAL MODULE INTERFACES

A1 | Bare PCB + hand soldering | Existing assembled modules remain external







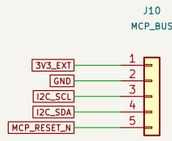


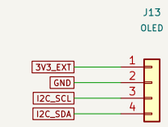


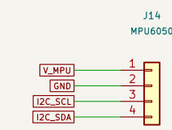


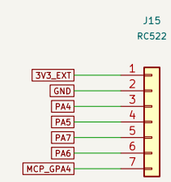


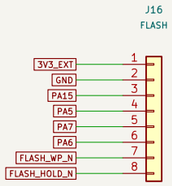


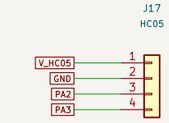


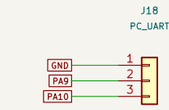


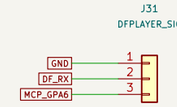


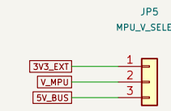


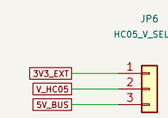


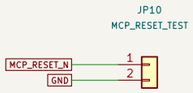












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

Sheet: /04 03 Serial module interfaces/
File: SmartBin_A_03_01.kicad_sch

Title: SmartBin A / 03 Serial module interfaces

Size: A3	Date: 2026-10-12	**Rev: A1**
KiCad E.D.A. 10.0.6		Id: 4/6

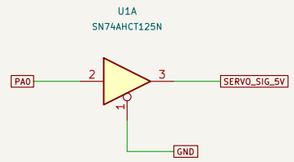
04 BUFFERED OUTPUTS AND ACTUATORS

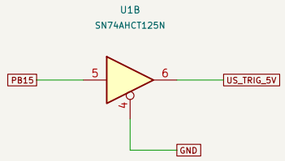
A1 | Bare PCB + hand soldering | Existing assembled modules remain external

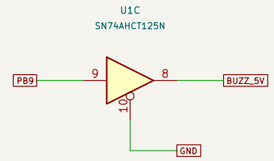


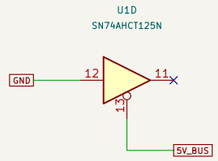


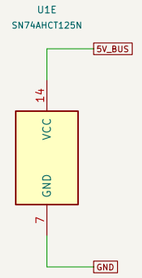


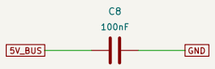














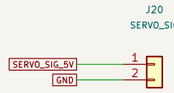


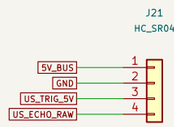


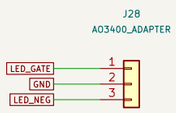




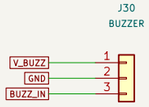


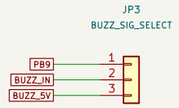


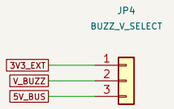












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

Sheet: /05 04 Buffered outputs and actuators/
File: SmartBin_A_04_01.kicad_sch

Title: SmartBin A / 04 Buffered outputs and actuators

Size: A3	Date: 2026-10-12	**Rev: A1**
KiCad E.D.A. 10.0.6		Id: 5/6

05 SENSOR AND USER INPUTS

A1 | Bare PCB + hand soldering | Existing assembled modules remain external







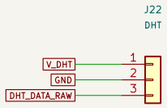


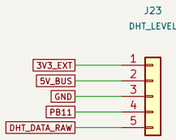


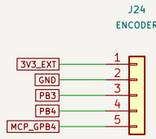


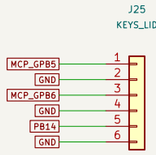


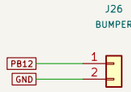


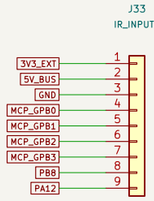


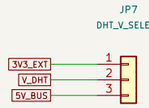


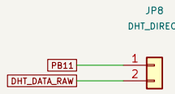












Prototype: verify polarity and module pin order before connection.
Existing external modules; cable connected; all named nets are global.
STM32 + RFID / DIY through-hole carrier

Sheet: /06 05 Sensor and user inputs/
File: SmartBin_A_05_01.kicad_sch

Title: SmartBin A / 05 Sensor and user inputs

Size: A3	Date: 2026-10-12	**Rev: A1**
KiCad E.D.A. 10.0.6		Id: 6/6